⊛ Dirac case NO

$$m_i = \sqrt{(\eta-1)(2i+r)\, b^2 \frac{\Lambda_H^2}{2\lambda N}} \quad \rightarrow \quad m_i^2 = \frac{1}{N}(2i+r)\, b^2 \frac{\Lambda_H^2}{2\lambda N}$$

$\eta - 1$ ↳ $1 + 1/N$

$$m_0^2 = \frac{1}{N}\cdot r \left(b^2 \frac{\Lambda_H^2}{2\lambda N}\right) s$$

↳ scale factor in the matrix

we sobstitute it $\longrightarrow$ with $s = \dfrac{m_0^2 \cdot N}{r}$

Therefore the eigenvalues are:

$$m_i^2 = \frac{(2i+r)}{r}\cdot m_0^2 \quad \rightarrow$$

There is no explicit scaling with N because the N suppression is inside m_0 which we fix

(In my diagonalization there's the factor $2^{\frac{1}{N-1}}$ but minimal)

The heavy mode scale as:

$$m_N^2 = \frac{1}{2}\left(1 + \frac{N}{\eta - 1}\right)(\eta-1)\, b^2 \frac{\Lambda_H^2}{2\lambda N}(2N+r)$$

↓ in terms of m_0

$$m_N^2 = \frac{1}{2}\left(1 + \frac{N}{\eta-1}\right)\frac{(2N+r)}{r} m_0^2 \sim \left(2N^2 + 2N^3\right)\frac{m_0^2}{2r}$$

$\eta - 1$: $1/N$

(or taking Manuel parametrization:

$$m_N^2 = N(N-1+r)\,(2N+r)\,\frac{\Lambda_H^2}{2\lambda N}$$

↳ $\sim N^3$

So in fact all eigenvalues are $m_i^2 = \frac{(2i+r)}{r} m_0^2$

and the last goes as $m_N^2 \sim N^3 m_0^2$.

⊛ Calculation of the effective mass:

$$m_\nu^2 = \sum_{i=1}^{3} \sum_{j=0}^{N} |U_{ei}|^2 |V_{0j}|^2 m_{ij}^2$$

$|V_{0j}|^2 \sim \frac{1}{N^2} \frac{(2i+r)}{4i^2}$ for small modes

$|V_{0N}|^2 \sim \frac{1}{N^2}$ for high modes

Contribution for small modes: $\underset{\text{many neutrinos}}{\underset{\downarrow}{N}} \cdot |V_{0j}|^2 \cdot m_{ij}^2 \sim N \cdot \frac{1}{N^2} 2i m_0^2$

$\sim$ at least goes as $\frac{1}{N}$

but

contribution from the high mode:

$\frac{1}{N^2} \cdot N^3 \sim N$ goes as N $\rightarrow$ the behaviour we see in the plot

⊛ The cut

The cut happens for energy order keV $\rightarrow 10^3$ eV $\rightarrow 10^6$ eV^2

For $N=100$ the contribution of the high mode is $\sim$

$100 \cdot m_0^2 \rightarrow 10^2 \cdot 10^{-4} \sim 10^{-2}$ (still within Katrin limit)
and the $m_N^2 \sim 10^6 \cdot 10^{-4} \sim 10^2$ eV^2 ✓

For $N=10^3$ $\rightarrow$ exclusion starts because the contribution is bigger than the katrin limit

$10^3 \cdot 10^{-4} \sim 10^{-1}$ (treshold of what is excluded without cut)

$m_H^2 = 10^9 \cdot 10^{-4} \sim 10^5$ eV^2

For $N=10^4$ $\rightarrow$ the cut starts (especially when r is small)
because $m_H^2 = 10^{12} \cdot 10^{-4} = 10^8$ eV^2 $> 10^6$ eV^2
↓
The heavy mode is cut out and the region is allowed again